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Paleomagnetism of the Schuller and Franspoort kimberlite pipes in South Africa and an improved Premier pole

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Abstract

Thirteen (out of 16) oriented kimberlite samples from the Schuller and Franspoort pipes in the Premier swarm have yielded consistent stable remanent vectors after detailed AF demagnetization. The calculated poles (Schuller, $n=8$, 33.4°N, 53.5°E, $\alpha_{95}=6.0$; Franspoort, $n=5$, 49.1°N, 48.8°E, $\alpha_{95}=7.5$) are similar to those previously determined for the Premier and National pipes. The four poles are somewhat streaked, but their mean: $n=4$, 30.0°N, 52.5°E, $\alpha_{95}=16.3$, is considered to be a better estimate of the pole position for the Kaapvaal Craton ~1200 Ma ago.

1. Introduction

Kimberlite pipes usually occur in "swarms" in which individual bodies represent a series of distinct, but evidently roughly coeval intrusions. Previous paleomagnetic studies conducted on the Premier and National kimberlite pipes in the Premier "swarm" (Fig. 1) by Jones (1968) and Hargraves and Onstott (1980), respectively, have yielded two paleomagnetic poles (see Table 3). As individual pipes do not seem to average out secular variation of the geomagnetic field, however (Hargraves, 1989), the average from several distinct pipes in the swarm would provide a more reliable pole position for the Kaapvaal Craton at the time of their emplacement. With that in mind, oriented samples were collected in 1987 from the only two additional pipes in the Premier swarm in which fresh rock was exposed: Schuller and Franspoort (see Fig. 1).

2. Age of Premier kimberlites

A minimum age for the Premier pipe of 1115 ± 15 Ma was obtained by Allsopp et al. (1976) who dated biotite (Rb–Sr) from a dolerite sill which cuts the pipe. Subsequent Rb–Sr, U–Pb and Sm–Nd studies of various kimberlite phases (see Table 1) have confirmed that the age of emplacement of Premier and National are around 1200 Ma. The pipes sampled in this study have not been dated directly, but in view of their geographic association, and the tendency for kimberlites to be intruded as swarms, they are assumed to be roughly coeval.

3. Sampling and measurement

At each of the pits which had been excavated at Schuller and Franspoort, eight cores were

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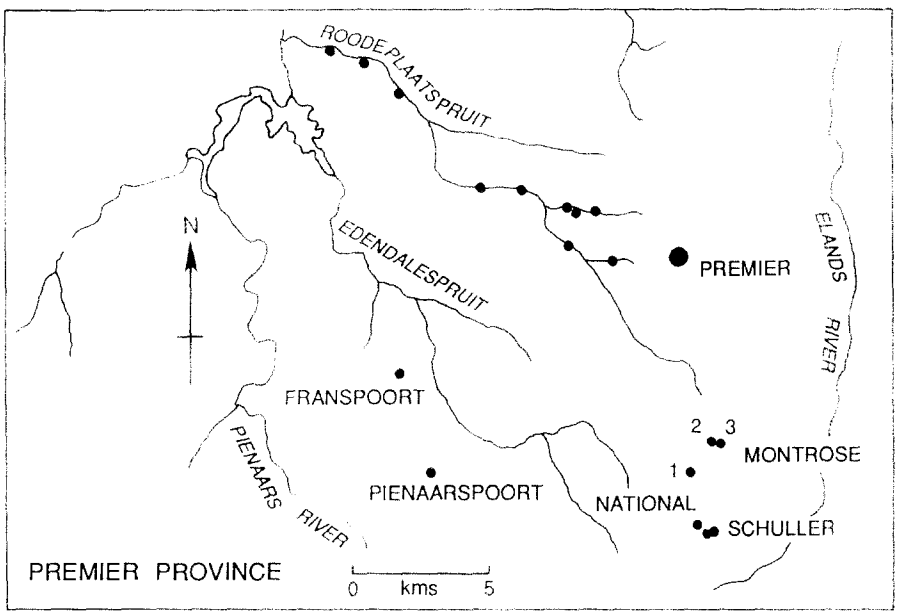


Fig. 1. Map showing location of pipes and other "shows" (dots) in the Premier kimberlite province, South Africa. The Premier mine is located at 25.4°S, 28.3°E.

Table 1
Age determinations for the Premier and National kimberlites

Age (Ma)	Isotopic system	Sample type	Reference
Premier			
1223	Rb–Sr	Phlogopite	Barrett and Allsopp (1973)
1174	Rb–Sr	Phlogopite	Allsopp and Kramers (1977)
1170	Rb–Sr	Phlogopite	Allsopp et al. (1979; quoted in Hargraves and Onstott, 1980)
1179±36	Rb–Sr	Diopside	Smith (1983) as cited in Phillips et al. (1989)
1202±72	Pb–Pb	Whole rock	Kramers and Smith (1983)
1180±30	Sm–Nd	Garnet	Jones (1984) as cited in Phillips et al. (1989)
1150±60	Sm–Nd	Garnet and clinopyroxene	Richardson (1986)
National			
1155±30	Rb–Sr	Phlogopite	Allsopp and Kramers (1977)
1140±16	Rb–Sr	Phlogopite	Allsopp et al. (1979) as quoted in Hargraves and Onstott (1980)

drilled in the field and oriented with a sun compass. The Schuller pit was actively being mined at the time of sampling (1987) and very fresh material was obtained. The Franspoort pit was shallower, and had been abandoned; the exposure here was extensively weathered but contained in-situ blocks of relatively fresh solid rock. None other of the known pipes in the Premier swarm (Fig. 1) had exposures suitable for sampling.

Remanence measurements were made on a Schonstedt DSM spinner, with AF demagneti-

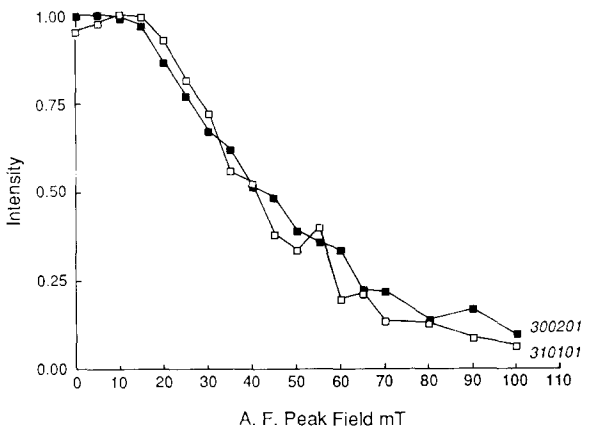


Fig. 2. Normalized AF demagnetized curves for representative samples from Schuller (*300201*) and Franspoort (*310101*).

zation on a custom built two-axis tumbler device (maximum field 120 mT); thermal demagnetization was performed in a Schonstedt TSD-1 furnace. All instruments are housed in a magnetically shielded room.

Specimens from all samples were AF demagnetized in seventeen steps to a 100 mT maximum. All cores responded well and similarly to this treatment, and representative normalized demagnetization curves for specimens from each site are shown in Fig. 2, indicating a median demagnetizing field (M.D.F.) of around 20 mT. Thermal demagnetization was performed on four specimens representative of both sites, in eighteen steps up to 660°C. In each, the bulk of the magnetization was unblocked between 500° and 600°, indicating magnetite as the primary carrier. This is consistent with optical observations and conclusions reached in previous paleomagnetic studies of kimberlites (Hargraves, 1989) that the stable remanence was carried by fine-grained magnetite produced during serpentinisation of the ferromagnesian minerals (penecontemporaneous with emplacement).

The results of the detailed AF demagnetization were analyzed using orthogonal vector plots (Zijderveld, 1967) (Fig. 3), and by linear regression (Kirschvink, 1980). A minimum of seven demagnetization steps were employed to obtain a valid vector with a mean angular deviation (MAD) less than 15° (usually less than 5°). Site averages were calculated using Fisher statistics and are presented in Table 2.

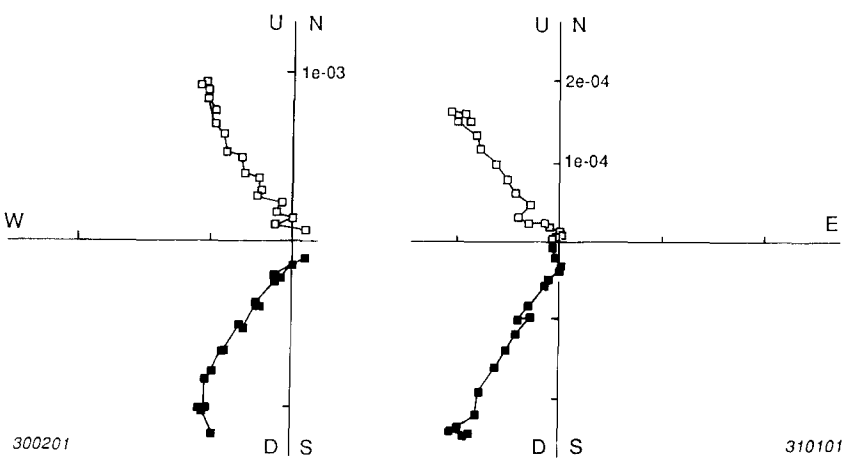


Fig. 3. Zijderveld diagrams illustrating vector migration during AF demagnetization of representative samples from Schuller (*300201*) and Franspoort (*310101*). After removal of a soft viscous component all samples showed a steady linear regression toward origin. Black squares = declination, Coliete squares = inclination.

Table 2
Paleomagnetic vectors

	N/n	D	I	k	α_{95}
Schuller, Site 30	8/8	202.9°	−44.4°	101	5.5
Franspoort, Site 31	8/5	193.4°	−24.3°	54	10.5

N/n samples drilled/used in analysis; D=declination; I=inclination; k=Fisher's precision parameter; α_{95}=cone of confidence.

Table 3
North Poles from Premier kimberlite swarm

	N	Lat.	Long.	α_{95}
Premier PMR	16	51.0°N	37.0°E	7.0
National NAT	11	21.0°N	65.0°E	7.0
Schuller SCH	8	33.4°N	53.5°E	6.0
Franspoort FRA	5	49.1°N	48.8°E	7.5
Average pole PAV	4	39.0°N	52.5°E	19.3

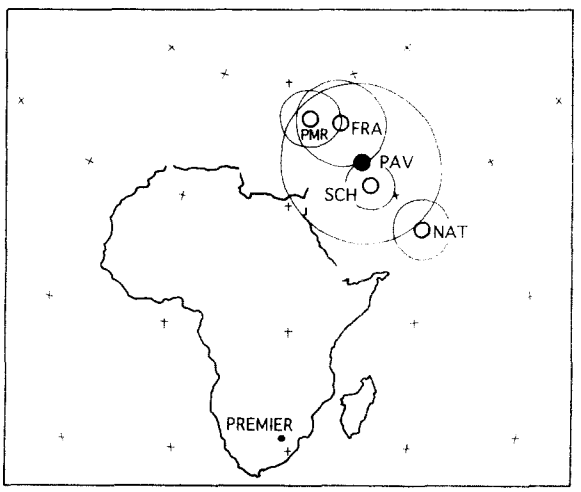


Fig. 4. Circles show the poles for Premier (*PMR*), National (*NAT*), Schuller (*SCH*), Franspoort (*FRA*); the dot shows the average pole of all four (*PAV*). See Table 3.

4. Discussion

Although the poles obtained in this study are similar to those published previously for Premier and National (and of the same polarity) the distribution of the four is noticeably linear (Fig. 4). Study of the Cretaceous kimberlites in South Africa (Hargraves, 1989), however, has demonstrated that individual pipes do not average secular variation, and the scatter of poles in Fig. 4 could plausibly be attributed to this cause. The dates reported for National and Premier appear somewhat different (Table 1) but given the notorious difficulty of getting exact dates for kimberlites (Allsopp et al., 1988; Hargraves, 1989) this difference may not be significant.

Despite these uncertainties, and their non-Fisherian distribution, we feel that the mean of these four poles is the best estimate for the pole position relative to the Kaapvaal Craton at the time of emplacement of the Premier kimberlite swarm, at ~1200 Ma.

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